Reaction Kinetics Sheet A

1) a) For a reaction $A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$:

$$t = \frac{1}{k_1 + k_{-1}} \ln\left(\frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}\right)$$

Show that this can be rearranged into the form:

$$[B] = \frac{[A]_0}{(k_{+1} + k_{-1})}(k_{+1} - k_{+1}\exp\{-(k_{+1} + k_{-1})t\})$$

$$t(k_1 + k_{-1}) = \ln\left(\frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}\right)$$

$$\exp[(k_1 + k_{-1})t] = \frac{k_1[A]_0}{k_1[A]_0 - (k_1 + k_{-1})[B]}$$

$$k_1[A]_0 - (k_1 + k_{-1})[B] = k_1[A]_0 \exp[-(k_1 + k_{-1})t]$$

$$-(k_1 + k_{-1})[B] = [A]_0(k_1 + k_1 \exp[-(k_1 + k_{-1})t])$$

$$[B] = \frac{[A]_0}{k_1 + k_{-1}}(k_1 + k_1 \exp[-(k_1 + k_{-1})t])$$

b) **(b)** If $[A] = \frac{[A]_0}{(k_{+1} + k_{-1})}(k_{-1} + k_{+1}\exp\{-(k_{+1} + k_{-1})t\})$, starting from time $t = 0$, by what proportion will [A] have decreased after a delay that lasts the length of the relaxation time, $\tau = 1/(k_{+1} + k_{-1})$?

Substitute $t = \tau = 1/k_1 + k_{-1}$

$$[A] = \frac{[A]_0}{k_1 + k_{-1}}(k_{-1} + k_1 \exp[-1])$$

$$\frac{[A]}{[A]_0} = \frac{k_{-1} + k_1/e}{k_{-1} + k_1}$$

(2) Following on from Equation (6.4) in the lecture handout, use partial fractions to show that:

$$\frac{1}{(p-[AB])(q-[AB])} = \frac{1}{(q-p)}\left(\frac{1}{(p-[AB])} - \frac{1}{(q-[AB])}\right)$$

$$\frac{1}{(p-[AB])(q-[AB])} = \frac{\alpha}{p-[AB]} + \frac{\beta}{q-[AB]}$$

$$\Rightarrow \quad 1 = \alpha(q-[AB]) + \beta(p-[AB]) = q\alpha - \alpha[AB] + \beta p - \beta[AB]$$

$$\Rightarrow \quad 1 = [AB](-\alpha-\beta) + q\alpha + p\beta$$

Compare coefficients $\quad -\alpha-\beta=0 \Rightarrow \alpha=-\beta$

$\therefore \quad q\alpha + p\beta = 1 \Rightarrow q\alpha - p\alpha = 1 \Rightarrow \alpha = \frac{1}{q-p}$

$$\therefore \quad \frac{1}{(p-[AB])(q-[AB])} = \frac{1/(q-p)}{p-[AB]} - \frac{1/(q-p)}{q-[AB]}$$

$$= \frac{1}{q-p}\left(\frac{1}{p-[AB]} - \frac{1}{q-[AB]}\right)$$

(3) In animal tissues, proteins are synthesized using only one optically active amino acid form, the *L*-enantiomer. Over time, however, *L*-amino acids slowly isomerise, producing measurable quantities of the *D*-enantiomer, via reactions that are not catalyzed by enzymes. If the rate constant for the isomerisation reaction is known, the age of an animal can be estimated using the ratio of the *D*- and *L*-forms of aspartic acid in samples of dentine taken from teeth.

(a) If the isomerisation reaction between the *L* and *D* enantiomers of aspartic acid obeys the chemical equilibrium equation:

$$L \underset{k}{\overset{k}{\rightleftharpoons}} D,$$ where the rate constants for the forward and backward directions are identical,

explain the significance of the following pair of coupled differential equations:

$$\frac{d[L]}{dt} = -k([L]-[D]) \quad \text{and} \quad \frac{d[D]}{dt} = +k([L]-[D])$$

$$\frac{d[L]}{dt} = k[D] - k[L] \qquad \frac{d[D]}{dt} = k[L] - k[D]$$

$D \xrightarrow{k} L \qquad L \xrightarrow{k} D$

(b) Determine a molar balance equation for the current concentrations of L and D in terms of their initial values.

Initial values @ $t=0$, $[L]_0 = [L]_0$, $[D]_0 = 0$

$$\therefore \quad [L]_0 = [L] + [D] \Rightarrow [L] = [L]_0 - [D]$$
$$\Rightarrow [D] = [L]_0 - [L]$$

(c) Show that t, the age of the sample, is given by:

$$t = -\frac{1}{2k} \ln\left[\frac{1 - [D]/[L]}{1 + [D]/[L]}\right]$$

Substituting from molar balance:

$$\frac{d[L]}{dt} = -k([L] - [L]_0 + [L]) = -k(2[L] - [L]_0)$$

$$\int_{[L]_0}^{[L]} \frac{d[L]}{2[L] - [L]_0} = \int_0^t -k\,dt = \left[\frac{1}{2}\ln(2[L] - [L]_0)\right]_{[L]_0}^{[L]} = -kt$$

$$\Rightarrow \frac{1}{2}\ln(2[L] - [L]_0) - \frac{1}{2}\ln([L]_0) = -kt$$

$$= \frac{1}{2}\ln\left(\frac{2[L] - [L]_0}{[L]_0}\right) = -kt$$

$$\Rightarrow t = \frac{-1}{2k}\ln\left(\frac{2[L] - [L]_0}{[L]_0}\right)$$

$$\Rightarrow t = \frac{-1}{2k}\ln\left(\frac{2[L] - [L] - [D]}{[L] + [D]}\right)$$

$$= \frac{-1}{2k}\ln\left(\frac{[L] - [D]}{[L] + [D]}\right) \quad \text{divide by } [L] \Rightarrow \frac{-1}{2k}\ln\left(\frac{1 - [D]/[L]}{1 + [D]/[L]}\right)$$

(d) Analysis of dentine from teeth of known age indicates that $k = 0.0020\ y^{-1}$. If the [D]/[L] ratio of a new sample is found to be 0.10, estimate its age.

$$t = \frac{-1}{0.0020\,yr^{-1}} \ln\left(\frac{1-0.1}{1+0.1}\right) = 50.2 \text{ years} \approx 50 \text{ years.}$$

Sheet B

(4) The equation that describes a reversible homo-dimerization reaction is:

$$A + A \underset{k_b}{\overset{k_a}{\rightleftharpoons}} B, \qquad 2A \underset{k_b}{\overset{k_a}{\rightleftharpoons}} B$$

(a) If the initial concentrations of A and B are $[A]_0$ and 0, respectively, determine a molar balance equation for the current concentrations of A and B.

$[A]_0 = [A] + 2[B]$ because B dissociates into 2A

(b) If $k_a = 1$, $k_b = 1$ and the initial concentrations of A and B are 1 and 0, respectively, determine the value of the equilibrium constant and calculate the equilibrium concentrations for A and B.

$$\frac{d[A]}{dt} = k_a[A]^2 - k_b[B] \qquad \text{(assuming mass action kinetics)}$$

$$= [A]^2 - [B]$$

At steady state: $1 = \frac{[B]_\infty}{[A]_\infty^2} = K_c$

$$\therefore 1 = \frac{[B]_\infty}{(\underset{\substack{\uparrow \\ 1}}{[A]_0} - 2[B])^2} = \frac{[B]_\infty}{1^2 - 4[B]_\infty + 4[B]_\infty^2}$$

$$\Rightarrow [B]_\infty = 1^2 - 4[B]_\infty + 4[B]_\infty^2$$

$$\Rightarrow 0 = 1^2 - 5[B]_\infty + 4[B]_\infty^2$$

Impossible!

$$0 = (1-[B]_\infty)(1-4[B]_\infty) \Rightarrow [B]_\infty = 1 \text{ or } 0.25$$

$$\Rightarrow [A]_\infty = 0.5$$

(c) Using the conditions specified in part **(b)**, solve the differential equation for [B] and determine the equilibrium concentration of B by examining the limit as time *t* tends towards infinity.

$$\frac{d[B]}{dt} = [A]^2 \cdot [B] = ([A]_0 - 2[B])^2 - [B]$$

$$= 1^2 - 5[B] + 4[B]^2$$

$$= (1-[B])(1-4[B])$$

$$\Rightarrow \frac{d[B]}{(1-[B])(1-4[B])} = dt$$ Need Partial fractions

$$\frac{1}{(1-[B])(1-4[B])} = \frac{\alpha}{1-[B]} + \frac{\beta}{1-4[B]}$$

$$\Rightarrow 1 = \alpha(1-4[B]) + \beta(1-[B])$$

$$= \alpha - 4\alpha[B] + \beta - \beta[B]$$

$$= [B](-\beta - 4\alpha) + (\alpha + \beta)$$

Comparing coefficients $-\beta - 4\alpha = 0$, $\alpha + \beta = 1$

$$\beta = -4\alpha \Rightarrow -3\alpha = 1 \Rightarrow \alpha = -1/3, \quad \beta = 4/3$$

$$\therefore \int_0^t dt = -\frac{1}{3}\int_0^{[B]} \frac{1}{1-[B]}\, d[B] + \frac{4}{3}\int_0^{[B]} \frac{1}{1-4[B]}\, d[B]$$

$$t = -\frac{1}{3}\left[-\ln(1-[B])\right]_0^{[B]} + \frac{4}{3}\left[-\frac{1}{4}\ln(1-4[B])\right]_0^{[B]}$$

$$= \frac{1}{3}\ln(1-[B]) - \frac{1}{3}\ln(1-4[B])$$

$$= \frac{1}{3}\ln\left(\frac{1-[B]}{1-4[B]}\right)$$

$$\Rightarrow e^{3t} = \frac{1-[B]}{1-4[B]} \Rightarrow e^{3t} - 4e^{3t}[B] = 1-[B]$$

$$\Rightarrow e^{3t} - 1 = 4e^{3t}[B] - [B]$$

$$= [B](4e^{3t} - 1)$$

$$\Rightarrow [B] = \frac{e^{3t}-1}{4e^{3t}-1}$$

as $t \to \infty$ $[B] \to 0.25$

(5) BILA398 is a potential drug being assessed as an inhibitor of the enzyme HIV protease. Its 1:1 complex with HIV protease has a dissociation constant of 6×10^{-12} M.

(a) Does this indicate tight binding or weak binding?

Very strong $K_d = \frac{[D]_\infty[E]_\infty}{[DE]_\infty} = 6\times10^{-12}M \ll\ll 1$, so $[DE] \gg\gg [D]$

$\parallel$

$6pM$

(b) If the initial concentrations of HIV protease and BILA398 are 1 pM and 3 pM, respectively, what is the final concentration of the complex? $\to [DE]_\infty$

$[D]_i = 3pM$ $[E]_i = 1pM$, $[E]_\infty = [E]_i - [DE]_\infty$

$[D]_\infty = [D]_i - [DE]_\infty$

$$\therefore K_d = \frac{([E]_i - [DE]_\infty)([D]_i - [DE]_\infty)}{[DE]_\infty} \quad x$$

$$\Rightarrow \quad 6 = \frac{(1-x)(3-x)}{x} = \frac{3-4x+x^2}{x}$$

$$\Rightarrow \quad 0 = 3 - 10x + x^2$$

$$\Rightarrow \quad x = 9.69\,pM \text{ or } \underline{0.31\,pM}$$

IMPOSSIBLE

(c) Is it sensible to progress BILA398 further as a drug candidate?

This drug binds 31% of protease. Not good.

(6) When considering the series of sequential unimolecular reactions $A \xrightarrow{k_a} B \xrightarrow{k_b} C$, the concentration of the intermediate B is given by the equation:

$$[B] = \frac{k_a[A]_0}{k_b - k_a}\left(\exp\{-k_a t\} - \exp\{-k_b t\}\right) = [B](t)$$

Determine an expression for the time t_{MAX} at which [B] reaches its largest value.

Maximise $[B](t)$, thus find $\frac{d[B]}{dt} = 0$

$$[B] = -J\exp(-k_a t) - J\exp(-k_b t)$$

$$\frac{d[B]}{dt} = -Jk_a\exp(k_a t) + Jk_b\exp(-k_b t)$$

$$= J\left(-k_a\exp(k_a t) + k_b\exp(-k_b t)\right) = 0$$

$$\Rightarrow k_b \exp(-k_b t) = k_a \exp(-k_a t)$$

$$\Rightarrow \frac{k_b}{k_a} \cdot \exp(-k_b t) = \exp(-k_a t)$$

$$\Rightarrow \ln\left(\frac{k_b}{k_a}\right) - k_b t = -k_a t$$

$$\Rightarrow \ln\left(\frac{k_b}{k_a}\right) = t(k_b - k_a)$$

$$\Rightarrow t = \frac{\ln(k_b/k_a)}{k_b - k_a} = \frac{\ln(k_a/k_b)}{k_a - k_b}$$

Sheet C

(7) If, in Michaelis-Menten enzyme kinetics, $K_M = (k_{-1} + k_{+2})/k_{+1}$, use inequalities to explain why on page 29 of the lecture handout (below Equation 10.16), the lecture handout states that any realistic choice of k_{+1} must exceed a threshold value of k_{+2}/K_M.

(**Hint:** What are the minimum possible values for k_{+2} and K_M?)

$$K_m k_1 = k_{-1} + k_2$$

$$k_1 = \frac{k_{-1}}{K_m} + \frac{k_2}{K_m}$$

So $k_1 > \frac{k_2}{K_m}$ constrained by the fact that all rate constants are positive.

(8) An extended version of the basic Michaelis-Menten scheme for an enzyme-catalysed reaction includes the unidirectional formation of an enzyme-product complex, followed by a unidirectional product release step:

$$E + S \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} ES \xrightarrow{k_{+2}} EP \xrightarrow{k_{+3}} E + P$$

(a) State the five differential equations that define this system and, assuming that the initial concentrations of ES, EP and P were zero, determine molar balance equations for the initial concentrations $[E]_0$ and $[S]_0$ in terms of the current concentrations of appropriate molecular species.

1 $$\frac{d[E]}{dt} = -k_1[E][S] + k_{-1}[ES] + k_3[EP]$$

2 $$\frac{d[S]}{dt} = \frac{d[E]}{dt} - k_3[EP]$$

3 $$\frac{d[ES]}{dt} = -k_2[ES] - k_{-1}[ES] + k_1[E][S]$$

4 $$\frac{d[EP]}{dt} = -k_3[EP] + k_2[ES]$$

5 $$\frac{d[P]}{dt} = k_3[EP]$$

$$[E]_0 = [E] + [ES] + [EP]$$

$$[S]_0 = [S] + [ES] + [EP] + [P]$$

(b) Apply the steady state approximation to [ES] and [EP] to derive expressions for [ES] and for [EP] in terms of [E], [S] and rate constants.

$$\frac{d[ES]}{dt} = -k_2[ES] - k_{-1}[ES] + k_1[E][S] = 0$$

$$\Rightarrow \quad 0 = k_1[E][S] - [ES](k_2 + k_{-1})$$

$$\Rightarrow \quad [ES] = \frac{k_1}{k_{-1} + k_2}[E][S]$$

$$\frac{d[EP]}{dt} = -k_3[EP] + k_2[ES] = 0$$

$$\Rightarrow [EP] = \frac{k_2}{k_3}[ES] = \frac{k_2}{k_3} \cdot \frac{k_1}{k_{-1}+k_2} \cdot [E][S]$$

(c) Use the molar balance equation for $[E]_0$ to determine an expression for [E] in terms of $[E]_0$, [S] and rate constants.

$$[E]_0 = [E] + [ES] + [EP]$$

$$[E] = [E]_0 - \frac{k_2}{k_3} \cdot \frac{k_1}{k_{-1}+k_2} \cdot [E][S] - \frac{k_1}{k_{-1}+k_2}[E][S]$$

$$[E] = [E]_0 - \overbrace{\frac{k_1}{k_{-1}+k_2}[E][S]}^{[ES]}\left(\frac{k_2}{k_3} + 1\right)$$

$$\Rightarrow [E]\left(\frac{k_1}{k_{-1}+k_2}[S]\left(\frac{k_2}{k_3}+1\right) + 1\right) = [E]_0$$

$$\Rightarrow [E] = [E]_0 \Big/ \left(\frac{k_1}{k_{-1}+k_2}[S]\left(\frac{k_2}{k_3}+1\right) + 1\right)$$

(d) Derive an expression for the rate of change of P in terms of $[E]_0$, [S] and rate constants.

$$\frac{d[P]}{dt} = k_3[EP] = \frac{k_2}{k_3} \cdot \frac{k_1}{k_{-1}+k_2} \cdot [E][S]$$

$$= k_2 \cdot \frac{k_1}{k_{-1}+k_2} \cdot [S][E]_0 \Big/ \left(\frac{k_1}{k_{-1}+k_2}[S]\left(\frac{k_2}{k_3}+1\right) + 1\right)$$

(e) Derive an expression for the initial rate of formation of P in terms of the initial concentrations of substrate and enzyme, an apparent Michaelis constant, K_M^{app}, and an apparent maximal rate, V_{MAX}^{app}.

At the start of the reaction $\frac{d[P]}{dt} = V_0$ and $[S] \approx [S]_0$

We know $V_0 = \frac{V_{max}^{app}[S]_0}{K_M^{app} + [S]_0}$

From prev question:

$$V_0 = \frac{k_2 \cdot \frac{k_1}{k_{-1}+k_2} \cdot [S][E]_0}{\left(\frac{k_1}{k_{-1}+k_2}[S]\left(\frac{k_2}{k_3}+1\right)+1\right)}$$

Multiply top and bottom by $1 / \frac{k_1}{k_{-1}+k_2}\left(\frac{k_2}{k_3}+1\right)$, sub $[S] = [S]_0$

$$V_0 = \frac{k_2 \cdot [S]_0[E]_0 \Big/ \left(k_2/k_3 + 1\right)}{[S]_0 + 1 \Big/ \frac{k_1}{k_{-1}+k_2}\left(\frac{k_2}{k_3}+1\right)}$$

Compare coefficients:

$$V_{max}^{app} = k_2[E]_0 \Big/ \left(k_2/k_3 + 1\right) = \frac{k_2[E]_0}{k_2 + k_3}$$

$$K_M^{app} = \frac{1}{\frac{k_1}{k_{-1}+k_2}\left(\frac{k_2}{k_3}+1\right)} = \frac{k_{-1}+k_2}{k_1\left(\frac{k_2}{k_3}\right)+k_1} = \frac{k_3(k_{-1}+k_2)}{k_1(k_2+k_3)}$$

(f) Use your definition of V_{MAX}^{app} to derive an expression for k_{CAT}. How do your expressions for K_M^{app} and k_{CAT} compare with the general definitions given on page 30 of the lecture notes?

$$k_{CAT} = \frac{V_{max}}{[E]_0} \Rightarrow k_{CAT} = \frac{k_2 k_3}{k_2+k_3}$$

(g) Under what circumstances could K_M^{app} be smaller than the dissociation constant of the enzyme-substrate complex, $K_{ES} = k_{-1}/k_{+1}$?

$$K_M^{app} = \frac{k_3(k_{-1}+k_2)}{k_1(k_2+k_3)} \qquad K_{ES} = \frac{k_{-1}}{k_1}$$

Under what condition is

$$\frac{k_3(k_{-1}+k_2)}{k_1(k_2+k_3)} < \frac{k_{-1}}{k_1}$$

Solve inequality

$$\Rightarrow \frac{k_3(k_1+k_2)}{(k_2+k_3)} < k_{-1}$$

$$\Rightarrow k_3(k_1+k_2) < k_{-1}(k_2+k_3)$$

$$\Rightarrow \cancel{k_3 k_1} + k_2 k_3 < k_{-1}k_2 + \cancel{k_3 k_{-1}}$$

$$\Rightarrow \cancel{k_2} k_3 < k_{-1}\cancel{k_2}$$

$$\Rightarrow k_3 < k_{-1}$$

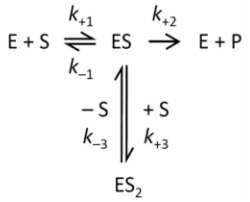
(b) Apply the steady state approximation to derive an expression for [ES] in terms of [E], [S] and rate constants.

extended reaction scheme considered here.

$$\frac{k_{cat}}{K_m^{app}} = \frac{\frac{k_2 k_3}{k_2+k_3}}{\frac{k_3(k_{-1}+k_2)}{k_1(k_2+k_3)}} = \frac{k_2 k_3 \cancel{(k_2+k_3)}}{k_3 k_3 (k_{-1}+k_2)\cancel{(k_2+k_3)}}$$

$$= \frac{k_2 k_1 \cancel{k_3}}{\cancel{k_3}(k_{-1}+k_2)} = \frac{k_1 k_2}{k_{-1}+k_2}$$

(9) A special case of uncompetitive inhibition occurs when a second substrate molecule binds to and inactivates the enzyme-substrate complex.



(a) Assuming that at time $t = 0$, $[S] = [S]_0$, $[E] = [E]_0$, and $[ES] = [ES_2] = [P] = 0$, apply the steady state approximation to derive an expression for $[ES_2]$ in terms of [S], [ES] and rate constants.

$$\frac{d[E]}{dt} = -k_1[E][S] + k_{-1}[ES]$$

$$\frac{d[S]}{dt} = -k_1[E][S] + k_{-1}[ES] + k_{-3}[ES_2] - k_3[S][ES]$$

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] - k_3[ES][S] + k_{-3}[ES_2]$$

$$\frac{d[ES_2]}{dt} = k_3[ES][E] - k_{-3}[ES_2]$$

$$\frac{d[P]}{dt} = k_2[ES]$$

$$\frac{d[ES_2]}{dt} = 0 \Rightarrow [ES_2] = \frac{k_3}{k_{-3}}[ES][S]$$

(b) Apply the steady state approximation to derive an expression for [ES] in terms of [E], [S] and rate constants.

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] - k_3[ES][S] + k_{-3}[ES_2] = 0$$

$$= k_1[E][S] - k_{-1}[ES] - k_2[ES] - \cancel{k_3[ES][S]} + \cancel{k_{-3}[ES][S]} = 0$$

$$= k_1[E][S] - [ES](k_{-1} + k_2)$$

$$\Rightarrow [ES] = \frac{k_1[E][S]}{k_{-1} + k_2}$$

(c) Use a molar balance equation to derive an expression for [E] in terms of $[E]_0$, [S] and rate constants.

$$[E]_0 = [E] + [ES] + [ES_2]$$

$$[E] = [E]_0 - \frac{k_1[E][S]}{k_{-1} + k_2} - \frac{k_3}{k_{-3}}[ES][S]$$

$$= [E]_0 - \frac{k_1[E][S]}{k_{-1} + k_2} - \frac{k_3}{k_{-3}} \frac{k_1[E][S]}{k_{-1} + k_2}[S]$$

$$\Rightarrow [E] + \frac{k_1[E][S]}{k_{-1} + k_2} + \frac{k_3}{k_{-3}} \frac{k_1[E][S]}{k_{-1} + k_2}[S] = [E]_0$$

$$\Rightarrow [E]\left(1 + \frac{k_1[S]}{k_{-1} + k_2} + \frac{k_3}{k_{-3}} \frac{k_1[S]^2}{k_{-1} + k_2}\right) = [E]_0$$

$$\Rightarrow [E]\left(1 + [S]\frac{k_1}{k_{-1} + k_2}\left(1 + \frac{k_3}{k_{-3}}[S]\right)\right) = [E]_0$$

$$\Rightarrow [E] = \frac{[E]_0}{\left(1 + [S]\frac{k_1}{k_{-1} + k_2}\left(1 + \frac{k_3}{k_{-3}}[S]\right)\right)}$$

(d) Derive an expression for the initial rate of formation of P in terms of $[E]_0$, $[S]_0$, the Michaelis constant ($K_M = (k_{-1} + k_{+2})/k_{+1}$), the maximal rate ($V_{MAX} = k_{+2}[E]_0$) and an inhibition constant ($K_I = k_{-3}/k_{+3}$).

$$\frac{d[P]}{dt} = V_0 = k_2[ES] = k_2 \underbrace{\frac{k_1[E][S]}{k_{-1}+k_2}}_{\text{from 9b}} = \frac{k_2[E][S]}{K_M}$$

$$\frac{k_1}{k_{-1}+k_2} = 1/K_M$$

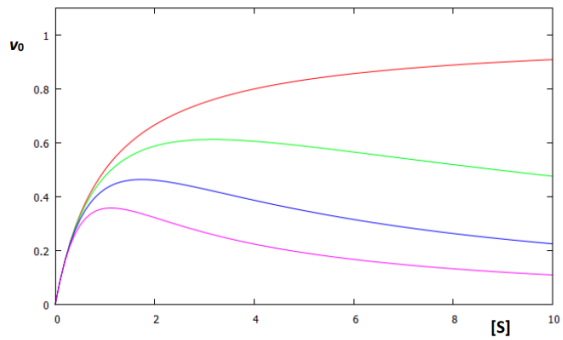
$$= \frac{k_2[S][E]_0}{K_M\left(1 + [S]\frac{k_1}{k_{-1}+k_2}\left(1 + \frac{k_3}{k_{-3}}[S]\right)\right)}$$

$$\frac{k_1}{k_{-1}+k_2} = 1/K_M \qquad \frac{k_3}{k_{-3}} = 1/K_I$$

$$= \frac{k_2[S][E]_0}{K_M\left(1 + \frac{[S]}{K_M}\left(1 + \frac{[S]}{K_I}\right)\right)}$$

$$= \frac{k_2[S][E]_0}{K_M + [S] + [S]^2/K_I}$$

(e) If $K_I = k_{-3}/k_{+3}$, and $K_I > (k_{-1} + k_2)/k_{+1} = K_M$, then the inhibitory site must bind to the substrate with weaker affinity than the active site. We would therefore expect a plot of activity as a function of substrate concentration to look something like this:



The figure above shows plots of $v_0/[E]_0$ for a series of increasing substrate concentrations, starting with K_M set to 1 and K_I set to infinity for the red curve, which has the rectangular hyperbola shape predicted by the normal Michaelis-Menten equation. The following curves have K_I values of 10 (green), 3 (blue) and 1 (magenta).

V_{MAX} can only be reached if $[S]_0/\{K_M + [S]_0 + [S]_0^2/K_I\} = 1$

i.e. if $[S]_0 = K_M + [S]_0 + [S]_0^2/K_I$

i.e. if $[S]_0^2/K_I + K_M = 0$

However, since $[S]_0$, K_I and K_M are all greater than 0, V_{MAX} can never be reached for this mode of inhibition.